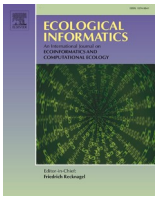
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Multi-modal, interrelated navigation in migratory birds: A data mining study

Ali Moayedi [a,*], Jed A. Long [b], Andrea Kölzsch [c,d,e], Helmut Kruckenberg [e], Fernando Benitez-Paez [a], Urška Demšar [a]

[a] School of Geography & Sustainable Development, University of St Andrews, St Andrews, Scotland, UK
[b] Department of Geography & Environment, Western University, London, Ontario, Canada
[c] Ecology Department, Radboud Institute for Biological and Environmental Sciences (RIBES), Radboud University, Nijmegen, Netherlands
[d] Department of Migration, Max Planck Institute of Animal Behaviour, Radolfzell, Germany
[e] Institute for Wetlands and Waterbird Research e.V. (IWWR), Verden (Aller), Germany

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ABSTRACT

Understanding how long-distance migratory birds navigate remains challenging, particularly in how they integrate multiple environmental cues. Traditional studies, primarily based on laboratory experiments and displacement studies, may not capture the complexity of navigation in the wild. In this study, we applied a data mining approach to investigate the navigational strategies of greater white-fronted geese (*Anser albifrons*) during their annual migrations between the Arctic and central Europe. We integrated a decade of tracking data from 117 individuals with high-resolution geomagnetic data (including solar-wind–induced variations), wind conditions, and a potential visual cue. Hierarchical cluster analysis revealed multi-modal and interrelated navigation strategies that flexibly adapted to environmental conditions such as wind, diurnal cycles, and flock-specific dynamics. Under favourable tailwinds, geese maintained stable headings with minimal changes in geomagnetic heading and apparent angle of geomagnetic inclination, consistent with both geomagnetic loxodrome and magnetoclinic routes, enabling efficient flights towards stopovers or simultaneously towards stopovers and final destinations. Geese also appeared to combine visual landmarks with geomagnetic information, adjusting their reliance on landmarks between day and night. Our findings highlight the complexity and adaptability of avian navigation and emphasise the role of multi-modal sensory integration and environmental context in shaping migratory decisions.

1. Introduction

Despite decades of research, the mechanisms and environmental cues guiding long-distance migratory birds continue to be a subject of ongoing scientific inquiry (Hagstrum, 2023; Lohmann et al., 2022; Mouritsen, 2018; Wallraff, 2004). This is particularly true for the process of *true navigation,* where birds utilise local information to navigate towards distant targets (Holland, 2014). Birds use a combination of global cues such as celestial cues, visual landmarks, olfactory information, infrasound, and Earth's magnetic field (Deutschlander and Beason, 2014; Gagliardo and Bingman, 2024; Hagstrum, 2000; Wallraff, 2005). However, it is still poorly understood how they integrate these cues within a complex and dynamic environment, translating them into effective navigational strategies and, ultimately, into movement decisions. This paper employs a new data-driven analysis approach to study navigational cue integration. We combined long-term tracking data from migratory birds with environmental data and used a data mining method (hierarchical clustering), to identify how patterns in navigational behaviour are related to the environmental conditions that birds experienced on their journeys. Although data-driven methods have been used to analyse animal movement (Dodge et al., 2014; Eikelboom et al., 2020; Wan et al., 2023), this is the first application of hierarchical clustering to examine how multiple cues might jointly shape real-time avian navigation.

The prevailing theoretical framework for understanding avian navigation is the *map and compass* model (Kramer, 1953). This model

* Corresponding author at: School of Geography & Sustainable Development, Irvine Building, University of St Andrews, North Street, KY16 9AL St Andrews, Scotland, UK.
E-mail address: am636@st-andrews.ac.uk (A. Moayedi).

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suggests that birds employ two distinct mechanisms: a map sense to determine their position relative to their destination (geographic positioning) and a compass sense to determine a desired heading (compass orientation). Within the map and compass framework, the Earth's magnetic field has been proposed as a potential source of both map and compass information (Lohmann et al., 2022). Nonetheless, researchers continue to debate whether geomagnetic cues alone are sufficient for the "map" sense or whether additional mechanisms—such as olfactory navigation (Bonadonna and Gagliardo, 2021; Gagliardo et al., 2021; Wallraff, 2004; Wallraff, 2005), infrasound (Hagstrum, 2000; Hagstrum, 2013; Patrick et al., 2021), or celestial compasses (Alerstam et al., 2001; Muheim et al., 2006)—are essential. Olfactory navigation, in particular, is supported by empirical evidence from studies on homing pigeons and wild birds (Bonadonna and Gagliardo, 2021; Gagliardo, 2013; Gagliardo et al., 2011; Gagliardo et al., 2021; Papi et al., 1971; Wallraff, 2005). However, global-scale data on olfactory and acoustic environments remain unavailable for long-term studies, whereas high-accuracy geomagnetic data are accessible worldwide via satellite sources (Benitez-Paez et al., 2021). Therefore, our study focuses on geomagnetic navigation while acknowledging the need for comprehensive global studies incorporating olfactory and acoustic cues for a full understanding of avian navigation.

The Earth's magnetic field is a complex and dynamic phenomenon, with the core field providing its foundation, the crustal field adding regional variations, and the external field, influenced by the solar wind, introducing dynamic fluctuations (Lanza and Meloni, 2006). An animal on the surface of the Earth or in the atmosphere experiences the cumulative effect of the core, crustal, and external fields—collectively referred to as the geomagnetic field. This geomagnetic field provides a global reference frame characterised by three primary variables: intensity (the strength of the geomagnetic field), inclination (the angle between the Earth's surface and the geomagnetic field lines), and declination (the angle between geomagnetic north and true north) (Fig. 1). These variables exhibit predictable variations across the planet under normal conditions, and it is widely hypothesised that birds possess biophysical capabilities enabling them to detect and use this information for orientation and map navigation (Wiltschko and Wiltschko, 2019).

Several theoretical models have been proposed to explain how birds might use the geomagnetic field for compass orientation. Two prominent models are the geomagnetic loxodrome and the magnetoclinic route. A geomagnetic loxodrome is a path of travel with a constant bearing relative to the geomagnetic north. This strategy is efficient for maintaining a steady course over long distances (Alerstam et al., 2001). A magnetoclinic route involves maintaining a constant apparent angle of geomagnetic inclination relative to the geomagnetic field. This is believed to help birds navigate by keeping a consistent orientation (Kiepenheuer, 1984).

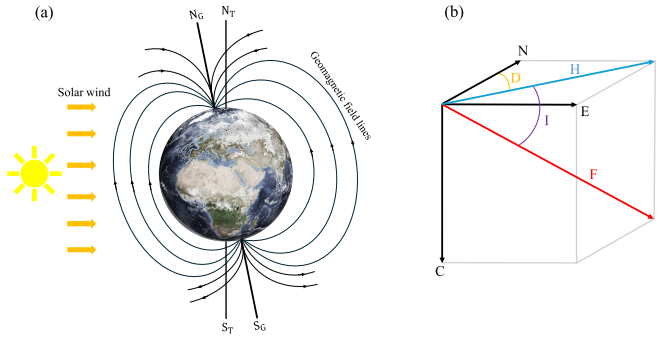


Fig. 1. Geomagnetic field and its components. (a) Schematic representation of geomagnetic field lines, extending from the geomagnetic south pole (S_G) to the geomagnetic north pole (N_G) and influenced by solar winds. True (or geographic) poles are denoted as S_T and N_T. (b) Representation of geomagnetic field components in the North-East-Centre (NEC) coordinate system: F represents the total geomagnetic field intensity, H is its horizontal component, I is the inclination (angle between F and H), and D is the declination (angle between geomagnetic north and true north).

Compass orientation is challenging because it can be disrupted by local environmental factors, particularly wind. Although favourable tailwinds can boost flight efficiency (Alerstam, 2011; Grönroos et al., 2012) and some birds time their departures accordingly (Åkesson and Hedenström, 2000; Kölzsch et al., 2016; Rüppel et al., 2023; Sjöberg et al., 2015), reliance on a single compass leaves birds vulnerable to drift (Chapman et al., 2011). To compensate, they must recalibrate their orientation by referencing a "map" sense, which may be geomagnetic, olfactory, or even infrasound-based (Chernetsov et al., 2008; Gagliardo et al., 2011; Hagstrum, 2000; Holland, 2014). While the map sense's exact mechanism remains contentious (Bonadonna and Gagliardo, 2021; Hagstrum, 2023; Wallraff, 2004; Wiltschko and Wiltschko, 2017), geomagnetic gradients—such as latitudinal variations in inclination and intensity—are hypothesised to provide birds with a positional framework (Dennis et al., 2007; Kishkinev et al., 2013; Lohmann et al., 2022; Wiltschko and Wiltschko, 2023). For a more detailed discussion of these debates, see the reviews by Able (2001) and Demšar et al. (2025). Additionally, localised geomagnetic anomalies, caused by geological features, might serve as navigational landmarks or signposts to confirm position and adjust course (Heyers et al., 2017; Lohmann, 1991; Wiltschko and Wiltschko, 2006).

Traditional investigations into avian navigation mechanisms have relied primarily on laboratory experiments and displacement studies, where individual birds are exposed to altered navigational cues to observe changes in their migration/movement behaviour (Chernetsov et al., 2008; Muheim et al., 2002; Wiltschko et al., 2006). While these controlled settings provide insights into the sensory and cognitive capabilities of birds, they offer a limited understanding of the real-time decision-making processes involved in long-distance migration within a complex and dynamic natural environment (Deutschlander and Beason, 2014). Furthermore, most studies focus on single navigational mechanisms, overlooking the possibility of a dynamic, multi-modal navigational system where birds flexibly switch between cues in response to prevailing conditions (Mouritsen et al., 2016).

Given the limitations of traditional approaches, there has been a growing interest in data-driven methods, such as data mining, to study avian movement. The increasing availability of high-resolution tracking data, combined with global-scale environmental datasets, has opened new opportunities to observe and analyse the behaviour of migratory birds during their journeys (Farley et al., 2018; Kays et al., 2022; Williams et al., 2020). Unlike conventional, hypothesis-driven research, which begins with a predefined question and collects data specifically to test it, data-driven approaches work in reverse, starting with large datasets to detect patterns, which then inform new research questions (Demšar et al., 2025). This shift from deductive to exploratory analysis enables researchers to uncover previously unrecognised relationships, forming abductive hypotheses based on emergent patterns rather than predefined assumptions (Miller and Goodchild, 2015).

Recent studies have begun to harness this potential, employing statistical movement models and machine learning techniques to investigate navigational strategies in migratory birds (Nicol et al., 2024; Wang, 2019). Agent-based models and correlated random walks, for instance, have been used to simulate and explore the potential role of geomagnetic cues in guiding migratory movements (Åkesson and Bianco, 2016, 2017; Zein et al., 2022). These approaches have already uncovered novel patterns, such as the influence of geomagnetic storms on bird behaviour, highlighting the promise of data-driven analysis in revealing unexpected insights (Benitez-Paez et al., 2021).

In this study, we adopt a data mining approach to investigate how greater white-fronted geese (*Anser albifrons*) navigate during migration. By integrating long-term GPS tracking data with detailed geomagnetic, wind, and visual cue information, we aim to identify when and where birds switch strategies in response to environmental cues. We

hypothesise that identified patterns will reflect established navigational strategies, such as geomagnetic loxodromes, magnetoclinic routes, or the use of geomagnetic anomalies as landmarks. By characterising these patterns, we aim to provide empirical evidence for diverse compass orientation strategies in free-flying birds and explore the role of geomagnetic cues in map-based navigation and compass recalibration. We also investigate the interplay between geomagnetic orientation and other sensory cues in shaping migratory decisions, examining how wind conditions influence navigation and the potential for adaptive flexibility in varying wind regimes. Our approach demonstrates the power of data-driven methods to uncover new patterns and behaviours, contributing to a more comprehensive picture of the multi-modal nature of avian migration.

2. Related work

Within movement ecology, analysing contextual variables is crucial for understanding movement decisions (Brum-Bastos et al., 2022), as movement is the primary way animals respond to their environment (Nathan et al., 2008). Various approaches have been used to link tracking data to environmental conditions. For example, Patterson et al. (2009) applied hidden Markov models (HMMs) to classify movement into predefined behavioural states (e.g., resident vs. migratory) using covariates like temperature anomalies. While effective, HMMs simplify complex navigation strategies in migratory birds by assuming fixed states and distributions. Similarly, Dodge et al. (2014) studied turkey vultures over a decade, linking large-scale migration to thermal uplift and vegetation, yet their work did not explore fine-scale navigational decisions. Eikelboom et al. (2020) introduced a machine learning framework to predict environmental variables (e.g., grass biomass) from high-resolution biologging data, capturing reactive movement patterns in dairy cows. However, their approach focuses on localised, short-term responses rather than proactive, goal-oriented navigation.

Recent advances in trajectory clustering have highlighted broader environmental associations. Wan et al. (2023) proposed a hierarchical framework integrating trajectory similarity measures (e.g., Fréchet distance) to cluster migration paths and compare environmental attributes across groups. While this reveals global patterns and individual movement preferences in relation to environmental factors, it does not address real-time decision-making or dynamic integration of cues. Similarly, Morten et al. (2025) identified six global marine flyways for pelagic seabirds using dynamic time warping (DTW) clustering of tracking data across ocean basins. Their work emphasises repeatable, large-scale routes aligned with wind-driven currents, providing a critical framework for transboundary seabird conservation. However, like Wan et al. (2023), their focus on ocean-basin connectivity does not resolve fine-scale, real-time navigational decisions or the interplay of competing cues during rapid environmental shifts.

In contrast to global trajectory clustering (Morten et al., 2025; Wan et al., 2023), our study targets the real-time, step-level integration of navigational cues. By integrating dynamic geomagnetic data, wind vectors, and potential landmarks at each GPS fix, we expose on-the-fly adjustments rather than broad-scale trends. Our approach not only identifies clusters of shared navigational tactics but also measures how closely these tactics guide birds to immediate or ultimate goals, thereby providing a more direct measure of navigational accuracy and goal-oriented decision-making in migratory birds.

3. Methods and data

3.1. GPS tracking data

We used high-resolution GPS tracking data from 117 adult greater white-fronted geese (*Anser albifrons*), ensuring that no two geese belonged to the same flock. These data, totalling over 6 million data points, were collected over a decade (2014–2024; see Data Availability section). The birds were caught in their breeding areas on Kolguyev Island (Russia) and at wintering sites in the Netherlands and Germany (Kölzsch et al., 2016; van Wijk et al., 2012). The geese were equipped with integrated GPS/GPRS neckband transmitters (35 g, madebytheo) that recorded location coordinates and height above mean sea level at intervals ranging from 1 to 30 min (median interval: 5 min). The data cover both spring and autumn migrations, enabling the study of seasonal variations in movement patterns and navigation strategies. Greater white-fronted geese migrate during both day and night (Kölzsch et al., 2016), making them an ideal species to explore how diurnal cycles influence navigation.

3.2. Tracking data preprocessing

Initial preprocessing involved the removal of data points with insufficient satellite fixes (<4) to ensure accurate geolocation. Outlier detection methods were employed to exclude points with anomalous locations and those exhibiting unrealistic speeds exceeding the known flight capabilities of the species, potentially indicative of GPS error (Hebblewhite and Haydon, 2010). To focus our analysis on the migratory behaviour of the North Sea population of greater white-fronted geese, we excluded individuals identified as having changed their population affiliation (Kölzsch et al., 2019). Finally, we removed tracks where geese did not complete the full migration between breeding and wintering sites, as these incomplete journeys could bias our analysis of migratory behaviour.

To isolate active migratory flight data from periods of foraging, moulting, or nesting, we implemented a multi-step filtering process. First, we identified stopover sites as locations where individuals stayed within a 30 km radius for more than 48 h (van Wijk et al., 2012). Similarly, nesting and moulting sites were defined as areas where geese stayed for more than 10 days within 2 km and more than 21 days within 30 km, respectively (Kölzsch et al., 2019). Data points within these areas were excluded. Next, we applied speed filters, removing data points with GPS-measured speeds below 10 m/s (Kölzsch et al., 2016) and calculated speeds (displacement/time) below 1.7 m/s (Zein et al., 2021), thereby retaining only data points likely representing active flight during migration.

To examine seasonal and diurnal variations in migratory behaviour, migration periods were defined as spring (1 March – 1 June) and autumn (15 August – 15 November) (Kölzsch et al., 2019). Data points were classified as nighttime when the solar depression angle was $\geq$18 degrees (end and start of astronomical twilight, when no indirect sunlight is visible and even faint celestial objects can be seen) and as daytime otherwise. This classification, implemented using the R package *maptools* (Bivand et al., 2022), resulted in four categories of migratory flight: spring daytime, spring nighttime, autumn daytime, and autumn nighttime.

3.3. Environmental data

We integrated geomagnetic, wind, elevation, and coastal proximity data to contextualise the geese's migration. For geomagnetic data, we used two complementary sources: the CHAOS-7 geomagnetic field model (Finlay et al., 2020) and the European Space Agency's Swarm satellite constellation (Friis-Christensen et al., 2006). The CHAOS-7 model provides baseline geomagnetic field values for each location and time, while Swarm measurements capture localised, dynamic variations caused by solar wind interactions with the Earth's magnetosphere. Following previous research (Zein et al., 2022), we employed MagGeo (Benitez-Paez et al., 2021), which integrates Swarm measurements with the CHAOS-7 model and accounts for solar wind effects through spatio-temporal interpolation (Iyer et al., 2022). This approach provides geomagnetic values at the birds' flight altitude with an accuracy of $\pm$21.6 nT (Benitez-Paez et al., 2021). From this integration, we calculated three geomagnetic parameters for each GPS data point: inclination

(I), declination (D), and total intensity (F), commonly used in studies of geomagnetic navigation (Hagstrum, 2023; Wiltschko and Wiltschko, 2019).

Wind conditions were extracted from the ERA5 reanalysis dataset at 100 m above ground level (Hersbach et al., 2020), a global resource produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and the Copernicus Climate Change Service, providing hourly U (zonal) and V (meridional) wind components at a 0.25° × 0.25° resolution (RMSE ≈ 1.35–3.69 m/s; Gualtieri, 2021). For each GPS fix, we interpolated wind components (U and V) using a spatio-temporal method inspired by the Env-DATA system (Dodge et al., 2013). Specifically, for each GPS data point, the two closest ERA5 raster datasets (one before, one after) were identified. The pixel containing each GPS point was located, and the values at the four corners and the centre were extracted. Distance-weighted interpolation was then applied to both raster datasets using these five reference points and inverse distance weights based on great circle distances. The two resulting values were linearly interpolated, weighted by the time difference between raster and GPS timestamps (Fig. 2). Incorporating the central pixel value provided an additional data point closer to the actual GPS fix location, which may help capture localised variations in wind conditions that corner values alone might not reflect.

In addition to geomagnetic and wind, we included elevation data from the ASTER Global Digital Elevation Model (horizontal resolution ≈ 30 m, vertical accuracy ±20 m; NASA/METI/AIST/Japan Spacesystems and U.S./Japan ASTER Science Team, 2019), enabling calculation of flight height. We also incorporated the Distance to the Nearest Coast dataset (*NASA Ocean Biology Processing Group*, 2009) to extract proximity to coastlines. For each GPS point, we extracted values using bilinear interpolation, which estimates values based on the weighted average of the four nearest grid cells.

After environmental annotation, we removed points with missing values, often caused by signal loss, sensor issues, or gaps in source data. To achieve consistent temporal resolution, we retained points with intervals between 4 and 10 min, discarding those outside this range. After completing all preprocessing steps, the dataset was reduced from over 6 million raw GPS points to 154,559 points for 104 individuals, optimised for analysing migratory behaviour (see Fig. S1 in the Supplementary Material). The final annotated dataset is publicly accessible via Zenodo (see Data Availability section). The dataset was further categorised into seasonal and time-of-day groups: spring-daytime (SD, n = 89,576), spring-nighttime (SN, n = 9284), autumn-daytime (AD, n = 46,549), and autumn-nighttime (AN, n = 9150).

3.4. Derivation of navigational and environmental cues

We derived six key navigational and environmental parameters detailed below. These cues, encompassing geomagnetic, wind, and visual inputs, could explain birds' orientation responses and movement strategies as location-based reactions to their environment.

3.4.1. Geomagnetic inclination change (ΔI)

This parameter measures the difference in geomagnetic inclination between two consecutive GPS points (Fig. 3a). Inclination, defined as the angle between geomagnetic field lines and the horizontal plane, may provide birds with directional information as they travel (Mouritsen, 2018; Wiltschko and Wiltschko, 2019). It is calculated as:

$$\Delta I_i = |I_{i+1} - I_i| \quad (1)$$

where ΔI_i represents the change of inclination between time points i and i + 1.

3.4.2. Geomagnetic heading change ($\Delta\psi_m$)

Geomagnetic heading refers to the bird's flight direction relative to geomagnetic north. This metric may help to understand how birds use geomagnetic information for navigation. We computed the geomagnetic heading (ψ_m) for each GPS data point using the bird's true heading (ψ) relative to the true north and the local geomagnetic declination (D) using the formula:

$$\psi_m = \psi + D \quad (2)$$

The change in geomagnetic heading ($\Delta\psi_m$) between consecutive GPS points is:

$$\Delta\psi_i = |\psi_{i+1} - \psi_i| \quad (3)$$

where $\Delta\psi_i$ is the change of geomagnetic heading between time points i and i + 1. This metric (Fig. 3b) may explain turning behaviour relative to geomagnetic north (Åkesson and Hedenström, 2007; Mouritsen, 2018).

3.4.3. Apparent inclination change (ΔI′)

The apparent angle of geomagnetic inclination, or apparent inclination (I′), represents the angle at which a bird perceives the geomagnetic field lines intersecting the horizontal plane relative to the direction of its movement (Fig. 4). While geomagnetic inclination (I) is a fixed geographical parameter at any given location, the apparent inclination (I′) is a dynamic parameter that varies depending on the bird's heading relative to geomagnetic north (ψ_m). Although there are alternative theories about how inclination can be sensed by birds (Mouritsen and Hore, 2012), the apparent inclination provides a mathematical model

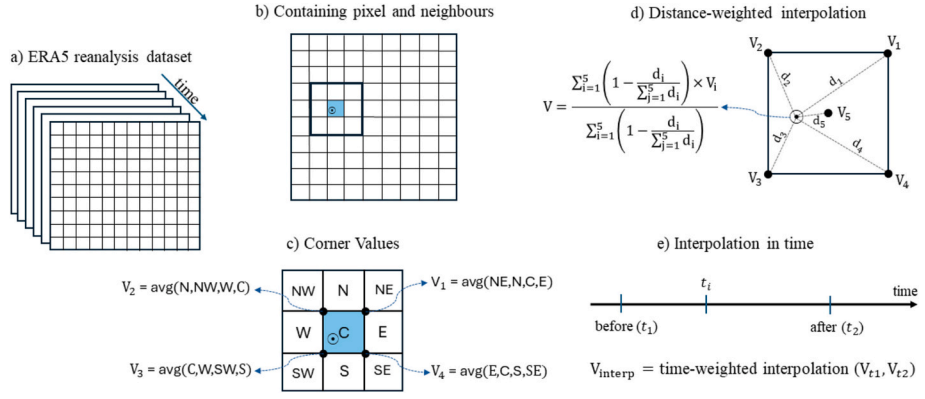


Fig. 2. Spatio-temporal interpolation process for annotating GPS tracking data with wind components from ERA5 reanalysis datasets. The figure presents a map view representation of the interpolation steps. (a) Temporal sequence of ERA5 raster datasets. (b) Identification of the pixel containing the GPS point and its neighbouring pixels. (c) Calculation of corner values using neighbouring pixel data. (d) Application of distance-weighted interpolation based on great circle distances to estimate values at the GPS location. (e) Temporal interpolation between two time-adjacent ERA5 datasets to derive final environmental variables at the exact GPS timestamp.

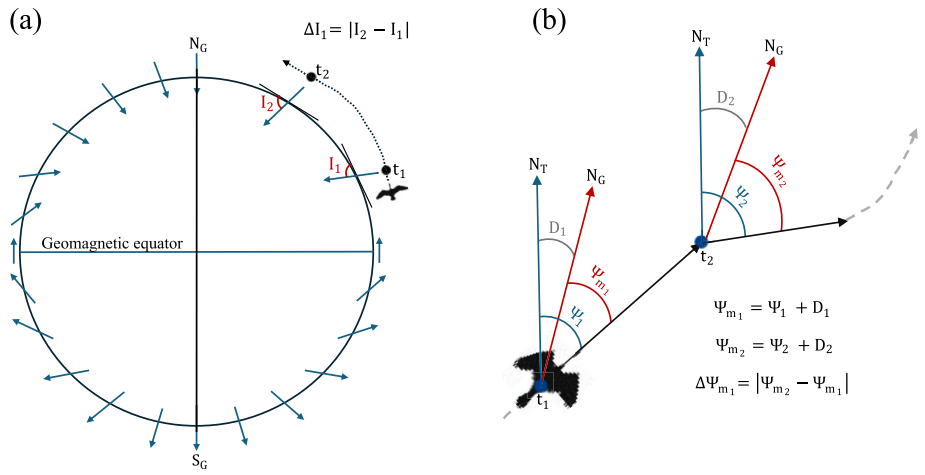


Fig. 3. Visualization of geomagnetic inclination change (ΔI) and geomagnetic heading change ($\Delta\psi_m$) along a bird's flight path. (a) illustrates the difference in geomagnetic inclination (ΔI) between two consecutive GPS points (t_1 and t_2). Blue arrows depict the geomagnetic field lines, converging towards the geomagnetic poles (N_G and S_G). (b) demonstrates the change in the geomagnetic heading (ψ_m) between consecutive points t_1 and t_2, calculated using the bird's true heading (ψ) and local geomagnetic declination (D). N_T and N_G indicate true north and geomagnetic north, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

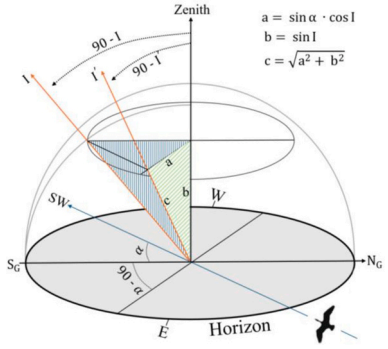


Fig. 4. Apparent inclination (I′) as perceived by a migrating bird. This figure depicts the apparent inclination (I′), which is influenced by the actual geomagnetic inclination (I) and the bird's flight direction (α) relative to the geomagnetic north (N_G). The green and blue shaded right-angle triangles illustrate the geometric relationship between the geomagnetic field components. The vertical component (b) represents the strength of the geomagnetic field perpendicular to the Earth's surface, while the horizontal component (a) aligns with the bird's heading. The hypotenuse (c) is the resultant vector, indicating the total perceived strength of the geomagnetic field as experienced by the bird. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

for describing a potential compass navigation strategy (Åkesson et al., 2021; McLaren et al., 2022). We calculated apparent inclination (I′) for each data point using the formula (Kiepenheuer, 1984):

$$\sin(I') = \frac{\sin(I)}{\sqrt{\sin(\psi_m)^2 \bullet \cos(I)^2 + \sin(I)^2}} \quad (4)$$

where I is inclination and ψ_m is geomagnetic heading. The change in apparent inclination (ΔI) between two points is:

$$\Delta I'_i = \left|I'_{i+1} - I'_i\right| \quad (5)$$

where $\Delta I'_i$ is the change of apparent inclination between time points i and i + 1. This dynamic parameter may influence directional adjustments and orientation during migration (Åkesson and Bianco, 2016; Muheim et al., 2018).

3.4.4. Distance to coastline (DtC)

This parameter measures the shortest distance from each GPS point to the nearest coastline. This feature could reveal the potential role of coastlines as guiding landmarks or navigational references during migration (Åkesson, 1993; Bruderer and Liechti, 1998).

3.4.5. Wind parameters (WS and CW)

Wind conditions during migration were characterised using wind support (WS) and crosswind (CW) (Fig. 5). Wind support, the component parallel to the bird's heading (ψ), is calculated as:

$$WS = U \bullet \cos(\psi) + V \bullet \sin(\psi) \quad (6)$$

Crosswind, the perpendicular component, is given by:

$$CW = V \bullet \cos(\psi) - U \bullet \sin(\psi) \quad (7)$$

Here, U and V are the interpolated zonal and meridional wind components at each GPS point, and ψ is the bird's heading. WS and CW significantly influence flight speed, energy expenditure, and route selection, providing insight into wind-related adaptations in migratory behaviour (Bradarić et al., 2020; Safi et al., 2013).

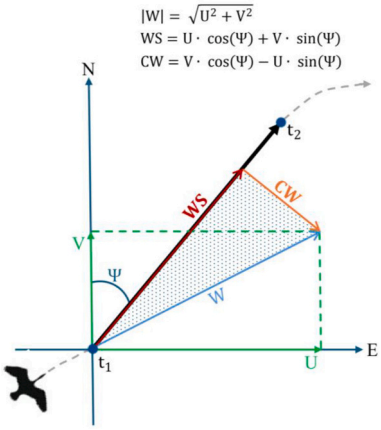


Fig. 5. Wind support (WS) and crosswind (CW) components relative to a bird's flight path. The U (zonal) and V (meridional) wind components are shown with their resultant wind vector (W). WS is the projection of W onto the bird's heading (ψ), and CW is perpendicular to the heading.

3.5. Cluster analysis

We applied agglomerative hierarchical cluster analysis to GPS fixes, using the navigational cues from Section 3.4 as cluster attributes, excluding spatial coordinates. This method aimed to reveal times and locations where bird movements correspond with similar navigational cues. We then hypothesised that by mapping and interpreting these clusters, we would be able to identify segments of migratory flight where different navigational environmental cues are being used.

Agglomerative hierarchical clustering (AHC) was chosen for its suitability for complex ecological data, as it does not require a predefined number of clusters and can effectively handle nested groupings often observed in natural systems (Murtagh and Contreras, 2017; Wan et al., 2023). Using Ward's linkage, we minimised within-cluster variance while maximising between-cluster variance, facilitating the identification of discrete behavioural modes (Murtagh and Legendre, 2014). Input data were *Z*-score standardised to ensure equal feature weighting, with outliers beyond ±5 standard deviations removed. Euclidean distance was used to measure dissimilarity between data points. The analysis was conducted independently for spring daytime, spring nighttime, autumn daytime, and autumn nighttime datasets to capture variations under different conditions.

Agglomerative Hierarchical Clustering (AHC) generates a dendrogram, a tree-like diagram that visualises how individual data points or small clusters merge into larger clusters at each step. Cluster determination involves analysing this dendrogram to decide where to cut it, which means selecting a level in the hierarchy where the clusters are defined. This step is critical because the number of clusters directly impacts the interpretation and biological relevance of the analysis. One common approach to determine the cut point is to use cluster validation indices such as the Silhouette index (Rousseeuw, 1987) and the Calinski-Harabasz index (Caliński and Harabasz, 1974). We chose to complement the above validation indices with a strategy that involved cutting the dendrogram at a height where the largest resulting cluster contained no more than 25 % of the total data points. This ensures a more balanced and granular clustering solution, preventing any single cluster from dominating the analysis and revealing potentially significant but smaller subgroups (Hennig, 2007; Stanberry et al., 2003).

To interpret the behavioural implications of the clusters, we calculated four movement parameters not included in the clustering process: movement speed, turning angle, and angular deviation from the direct path to both the next stopover and the final destination (Table 1; Fig. 6). This two-step approach, where clusters are first defined by specific attributes and then interpreted using additional behavioural metrics, ensures that interpretation remains unbiased by the initial clustering criteria, allowing a clearer link between environmental conditions and behavioural responses (Barraquand and Benhamou, 2008; Mose and Western, 2015; Wan et al., 2023).

Movement speed and turning angle are widely used metrics in movement ecology (Turchin, 1998). Turning angle reveals variations in flight path tortuosity (Estevez and Christman, 2006), while movement speed quantifies pace during migratory flight (Demšar et al., 2015). To avoid inaccuracies from instantaneous measurements, these metrics were calculated between consecutive GPS points using the *move* package in R (Kranstauber et al., 2018). Angular deviation, on the other hand, quantifies how closely birds align their headings with the shortest path to migratory targets, providing insight into navigation accuracy. We defined angular deviation from the direct path as the angular difference between the bird's heading at a given GPS point and the bearing to either its final destination (σ_D) or the centre of its next stopover area (σ_S).

We further examined cluster membership to assess whether the identified patterns were general across the population or influenced by specific individuals. To do so, we analysed the distribution of individuals within clusters and applied the chi-square test to detect any over-representation of individual-specific behaviour.

Table 1
Summary of movement parameters for cluster interpretation.

Movement Parameter	Description	Purpose
V	Calculated movement speed (m/s) between consecutive GPS fixes	Quantifies movement pace
θ	Absolute turning angle (°) between consecutive GPS fixes	Measures tortuosity and directional change
σ_D	Deviation (°) from the great-circle route to the final destination	Measures navigational precision towards the ultimate goal
σ_S	Deviation (°) from the great-circle route to the next stopover centre	Measures navigational precision towards the intermediate goal

4. Results

4.1. Migration patterns and wind conditions

The migratory routes of the greater white-fronted geese were generally situated near the Baltic-Barents-Kara coastal migration corridor, with this coastal preference being evident in both spring and autumn, though more pronounced in autumn (Fig. 7). Some individuals deviated from this route during spring, venturing further inland over eastern Europe. Autumn migrations, while shorter in duration, covered greater overall distances than spring migrations due to additional travel to moulting sites by failed and non-breeders (Kölzsch et al., 2019; Table 2). Autumn migrations were also characterised by a more direct flight path with lower heading variability compared to spring. Conversely, spring migrations were marked by greater variability in headings, with some individuals deviating significantly from the most direct path. Wind conditions along the migratory route also influenced these patterns. Tailwinds were prevalent in both seasons, comprising 65 % of the observed wind conditions during spring and 67 % in autumn. During spring migrations, geese experienced a significant 105 % increase in wind support at night compared to daytime. In contrast, wind support during autumn migrations remained relatively consistent between day and night.

4.2. Hierarchical clustering

The hierarchical clustering results were visualised through dendrograms, illustrating the grouping of GPS points based on their similarity in navigational and environmental attributes (Fig. 8 for daytime migrations; Fig. S2 for autumn migrations in the Supplementary Material). These dendrograms revealed the progressive grouping of individual GPS points based on their similarity in the attribute space, with each branching point representing a fusion of clusters. Initial assessment using the Silhouette and Calinski-Harabasz indices suggested two primary clusters for each migratory period, revealing a fundamental division in migratory behaviour. This primary split consistently separated a smaller proportion of data points characterised by notably high changes in apparent inclination ($\Delta I'$) and geomagnetic heading ($\Delta\psi_m$) from the remaining majority.

In the spring daytime migration dendrogram (Fig. 8a), the initial split isolated cluster SD1 (21 %), characterised by the highest changes in apparent inclination ($\Delta I'$) and geomagnetic heading ($\Delta\psi_m$). The larger branch (79 % of the data) is divided further into subgroups, including SD2 (4 %) with high changes in geomagnetic inclination (ΔI), SD3 (14 %) representing inland deviations, and SD4–SD6, which are differentiated by varying wind conditions, such as headwinds, tailwinds, and crosswinds. A similar pattern was observed in the spring nighttime dendrogram (Fig. 8b), where the smaller branch (SN1 and SN2, 11 % combined) showed the highest $\Delta I'$ and $\Delta\psi_m$. Within the larger branch (89 %), SN6 (5 %) are distinguished by high changes in geomagnetic inclination (ΔI), while SN4 and SN5 (25 % combined) include inland

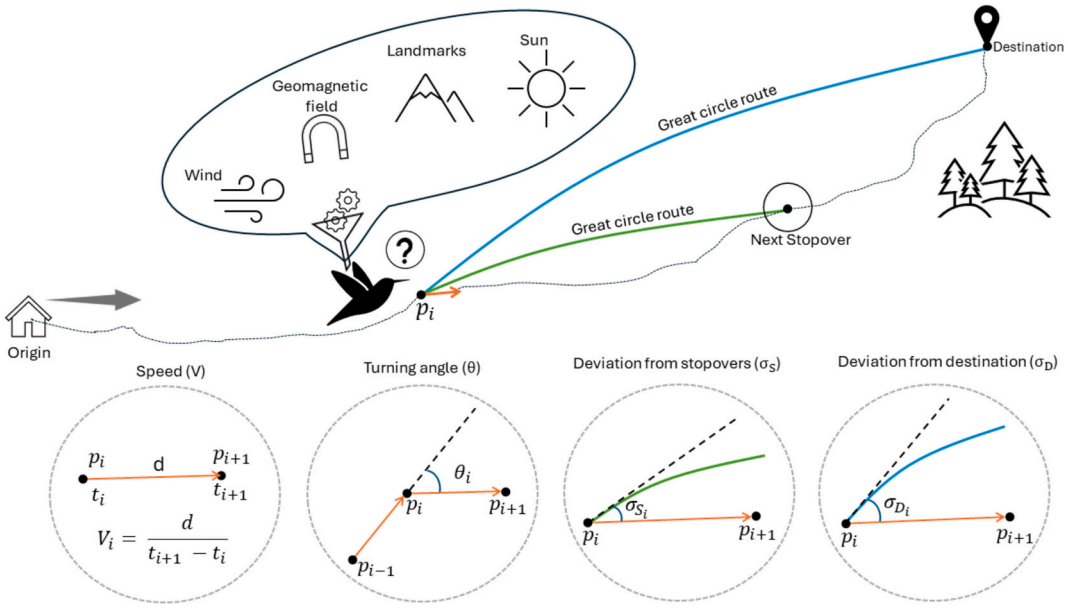


Fig. 6. Schematic representation of the movement parameters used in cluster interpretation. The geese integrate various sensory inputs to determine their direction at each GPS point (p_i). The figure illustrates how speed (V), turning angle (θ), and deviations from direct routes to the next stopover (σ_S) and final destination (σ_D) are calculated. The dashed line represents the actual migration path from origin to destination. The green and blue lines show the great circle routes (shortest paths) connecting p_i to the next stopover and the final destination, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

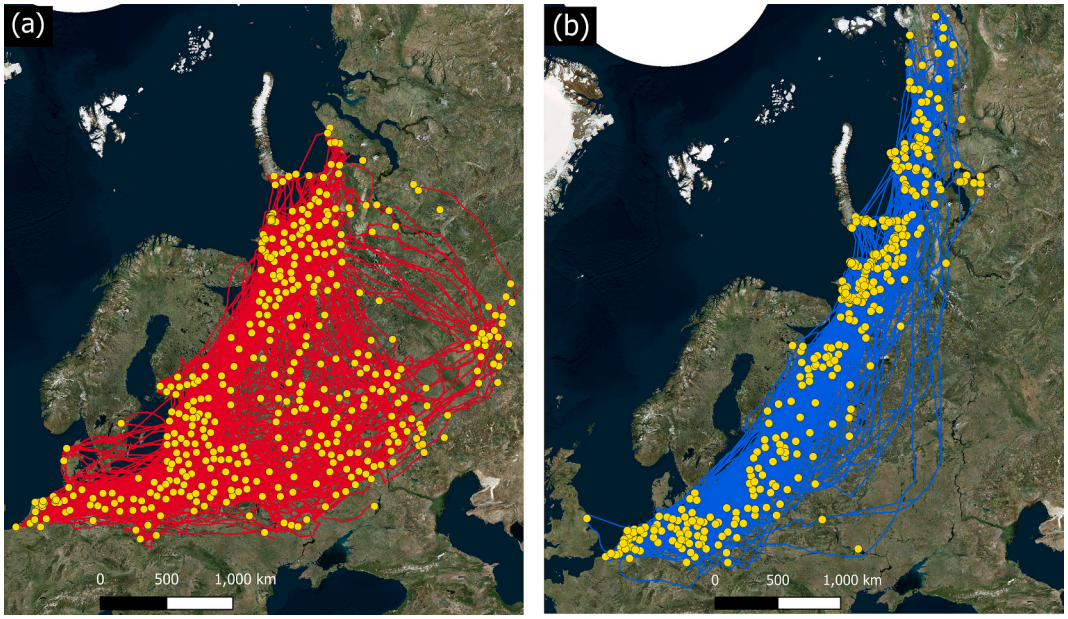


Fig. 7. Migratory routes and stopovers of the North Sea population of greater white-fronted geese. (a) Spring migration. (b) Autumn migration. Yellow dots indicate identified stopover sites along the migratory routes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

deviations. The remaining clusters (59 %) exhibited moderate geomagnetic changes and variable wind conditions. This pattern was mirrored in the autumn migrations (Fig. S2), where the minority of data fell within the first branch with high ΔI' and $\Delta\psi_m$ (4 % for daytime and 5 % for nighttime), while the rest resided in a large, non-homogeneous branch with subgroups exhibiting inland routes, high ΔI, and diverse strategies influenced by various geomagnetic cues and wind conditions.

While the two-cluster solution revealed a clear behavioural split, it was dominated by a large, heterogeneous cluster, limiting deeper interpretation. To achieve a more granular understanding of the migratory strategies and uncover additional patterns, a 25 % data threshold was applied, as described in the method section. This adjustment resulted in a more detailed classification, yielding 6 clusters for spring daytime (SD), 9 for spring nighttime (SN), 11 for autumn daytime (AD), and 11 for autumn nighttime (AN).

Clusters were defined by their navigational cues, including change in apparent inclination (ΔI'), change in geomagnetic inclination (ΔI), change in geomagnetic heading ($\Delta\psi_m$), distance to the coast (DtC), wind support (WS), and crosswind (CW). Each cluster was labelled based on the dominant features in its cue distributions, highlighting the dominant

Table 2
Summary statistics of migration characteristics for spring and autumn migrations.

Statistic description	Spring mean ± sd	Autumn mean ± sd
Overall ground speeds (m/s)	18.51 ± 5.5	18.03 ± 5.2
Total distance travelled (km)	4115 ± 1455	3854 ± 925
Total duration (days)	61.13 ± 19.9	53.21 ± 19.7
Great circle distance (km)	2498 ± 587	3131 ± 725
Overall headings (degrees)	115.92 ± 101.08	232.70 ± 46.3

behaviours associated with each cluster (Fig. 9, Figs. S3–9 in the Supplementary Material).

Analysis of cluster membership revealed uneven individual distributions across clusters (chi-squared test, $p < 0.001$), indicating that certain geese contributed disproportionately to specific clusters. During the spring daytime period, most clusters had median proportions ranging from 0.61 % to 0.82 %, below the expected 0.96 % for an even distribution among 104 individuals. This disparity was most pronounced in Cluster SD3: Inland, which exhibited the lowest median proportion (0.55 %) and the highest maximum proportion (11.4 %). Further details are provided in the Supplementary Material.

Wind conditions strongly influenced cluster formation, revealing different strategies under varying wind regimes. High crosswinds were a notable feature of autumn migrations, particularly in cluster AD9 (12 % of data, median 8.4 m/s), whereas crosswinds in spring were milder, with a median of 6.8 m/s in SD4. Headwinds appeared across all periods, strongest during autumn nights (AN11, median − 4 m/s), and moderate during spring days (SD6: 24 %, median − 1.6) and autumn days (SD10: 10 %, median − 1.7). Tailwinds were often associated with low changes in apparent inclination (ΔI') and geomagnetic heading ($\Delta\psi_m$). For instance, the "Low ΔI' and $\Delta\psi_m$, Tailwinds" cluster comprised 24 % of spring-day data (SD5, median 4.9 m/s), 10 % of autumn-day data (AD7, median 8.1 m/s), and 23 % of autumn-night data (SN9, median 6.8 m/s).

To visualise the spatial distribution of each environment-based cluster, we plotted every GPS fix with its cluster label (Fig. 10; Figs. S10-S12). Since clustering used navigational cues instead of geographic location, most clusters appear throughout the entire migratory route, showing that similar environmental conditions can arise in widely separated areas. The only exception is the Inland cluster, which mainly comprises points located far from the coastline. Nevertheless, points at great distances from the coast may still be assigned to other clusters if their navigational and environmental attributes exert a stronger influence than coastal proximity.

4.3. Interpretation of clusters using movement parameters

Building on the previous section, which characterised clusters based on navigational cues, we examined movement parameters not included in the clustering process. This analysis provided insights into how clusters corresponded to specific movement behaviours, such as pace and direction.

We analysed the distribution of movement speed (V) and turning angle (θ) within clusters across the four migratory periods (Fig. 11; Figs. S13–15). Clusters with high changes in apparent inclination (ΔI') and geomagnetic heading ($\Delta\psi_m$), such as the "High ΔI' and $\Delta\psi_m$" clusters, consistently exhibited lower speeds and higher turning angles, suggesting exploratory behaviour or frequent course corrections across all migratory periods. In contrast, clusters associated with tailwinds and minimal changes in I' and ψ_m, such as "Low ΔI' and $\Delta\psi_m$, Tailwinds" clusters, showed the highest speeds and lowest turning angles, reflecting fast, direct flights under favourable wind conditions. Both patterns were present throughout seasons and times of day.

To assess migratory navigation accuracy and efficiency, we analysed deviation of geese headings from the most direct path (great-circle) to both destinations (σ_D) and stopovers (σ_S) (Fig. 12; Figs. S16–18). We found significant differences in these deviations across clusters in all migratory periods (Kruskal-Wallis test, $p < 0.001$), suggesting that cluster-specific factors influence navigation. Clusters with high changes in geomagnetic values (e.g., "Highest ΔI' & $\Delta\psi_m$") exhibited larger deviations from both stopovers and destinations, reflecting greater navigational challenges. In contrast, clusters with minimal changes in apparent inclination and geomagnetic heading consistently showed lower deviations, suggesting more direct and goal-oriented flight paths.

During spring daytime, the "Low ΔI' & $\Delta\psi_m$, Tailwinds" cluster exhibited the lowest deviations from both destinations and stopovers, with significant differences from other clusters (Dunn's test, $p < 0.05$). On autumn days (Fig. S17), the same cluster showed the lowest deviations from stopovers, also with significant differences from other clusters (Dunn's test, $p < 0.05$), while deviations from destinations were not statistically distinct. At night, similar trends were observed. During spring nights (Fig. S16), the "Low ΔI' & $\Delta\psi_m$, Tailwinds" cluster had the lowest deviation from stopovers, with significant differences from most clusters but not from the "Low ΔI' & $\Delta\psi_m$" cluster. For deviations from destinations, no cluster showed a significant advantage. On autumn nights (Fig. S18), the "Low ΔI', Tailwinds" cluster had the lowest deviations, but these were not statistically significant.

Deviations from destinations (σ_D) were greater in spring than in autumn, with nighttime migrations generally showing larger deviations. This pattern was particularly pronounced for σ_D, suggesting that diurnal cues might have aided in maintaining a more direct course towards destinations. In spring, the increase in σ_D at night coincided with a

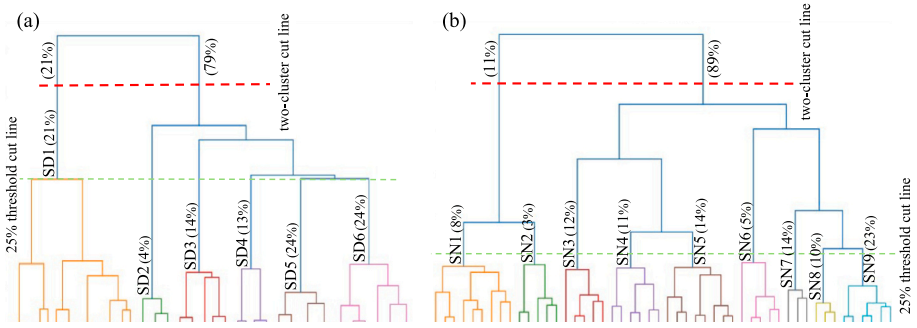


Fig. 8. Dendrograms of hierarchical clustering for migratory behaviours in greater white-fronted geese during (a) spring daytime and (b) spring nighttime migrations. The red dashed line marks the initial two-cluster division, with the dominant clusters containing 79 % (day) and 89 % (night) of the data. The green dashed line indicates the refined cut, resulting in 6 clusters for daytime and 9 for nighttime, with clusters colour-coded and percentages representing the proportion of data in each group. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

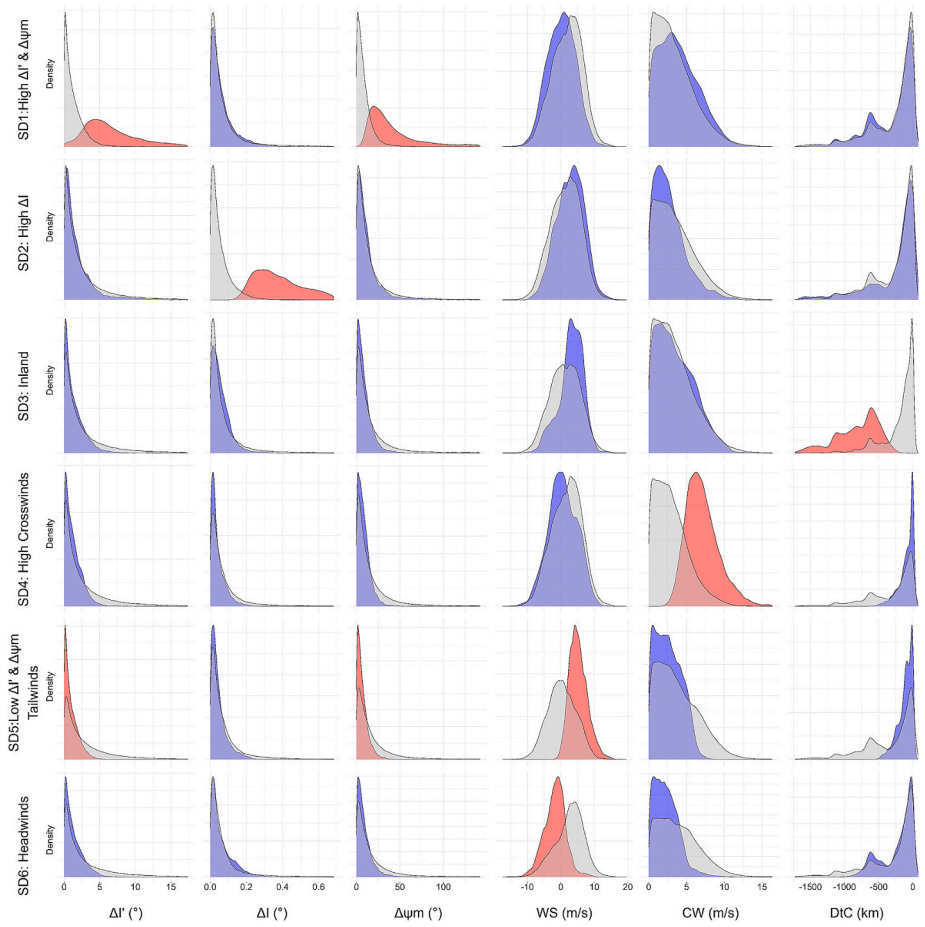


Fig. 9. Distribution of navigational cues across clusters for spring daytime (SD) migration. Columns represent navigational cues, and rows correspond to distinct clusters. Grey density curves show the overall distribution of each cue across all data, excluding the focal cluster. Blue and red density curves represent the variable distributions within the focal cluster, with blue indicating alignment with the overall dataset (grey) and red highlighting distinct, cluster-specific patterns. Each cluster is labelled according to the key navigational cues that most prominently differentiate it. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

reduction in σ_S, potentially indicating a trade-off between accurately reaching stopover sites and maintaining an overall migratory direction.

To supplement our analysis, we examined the circular distributions of headings (Fig. 13; S19–22), revealing directional preferences during migratory flights. Clusters with high ΔI' and $\Delta\psi_m$ exhibited broader heading dispersion, indicative of exploratory behaviour, while clusters with Low ΔI' and $\Delta\psi_m$ maintained focused, goal-oriented headings aligned with migratory routes. A clear distinction emerged between day and night, with nighttime flights revealing stronger longitudinal components and greater deviations from expected migratory directions. In spring, eastward headings dominated nighttime movements, contrasting with the north-easterly paths typical of daytime flights. Similarly, autumn nighttime flights shifted westward, diverging from the south-westerly daytime trend.

5. Discussion

Our findings indicate that greater white-fronted geese navigate in a flexible and context-dependent manner, integrating multiple cues rather than relying on a single strategy. By applying hierarchical clustering to long-term, high-resolution tracking data, we uncovered patterns that remain hidden when examining cues independently. A key observation was the frequent occurrence of "turning point" clusters—characterised by notable changes in apparent inclination (ΔI') and geomagnetic heading ($\Delta\psi_m$)—during spring migrations, a season in which resource constraints and migration goals may necessitate more complex navigational adjustments (Kölzsch et al., 2016). In autumn, these turning points were much less common, as geese travelled more directly, reflecting different environmental conditions and a reduced need for mid-flight course changes. Notably, the eastward divergence observed in spring tracks likely reflects increased foraging efforts aimed at resource accumulation for optimal breeding, whereas autumn migrations are more direct due to the abundance of food resources at both departure and arrival (Kölzsch et al., 2016; van Wijk et al., 2012). Diurnal cycles added another layer of complexity, with nighttime flights more closely aligned with wind direction and prioritising intermediate stopovers over distant destinations. Individual variations in cluster membership highlighted that not all geese responded alike, highlighting the adaptive and individualised nature of their navigational decisions. Together, these results highlight how exploratory data mining can uncover differences in behaviours and provide a foundation for future hypothesis-driven studies to clarify the underlying causes.

Among these flexible navigation strategies, clusters marked by minimal changes in apparent inclination (ΔI') and geomagnetic heading ($\Delta\psi_m$), especially under tailwind conditions, stood out for their navigational efficiency. In particular, the "Low ΔI' & $\Delta\psi_m$, Tailwinds" cluster demonstrated the highest alignment with the shortest migratory paths. These clusters consistently exhibited stable headings, minimal turning angles, and high ground speeds, suggesting that geese were maintaining a consistent and direct course during migration. Such navigation efficiency may represent a form of compass orientation, potentially resembling geomagnetic loxodrome and magnetoclinic routes (Åkesson

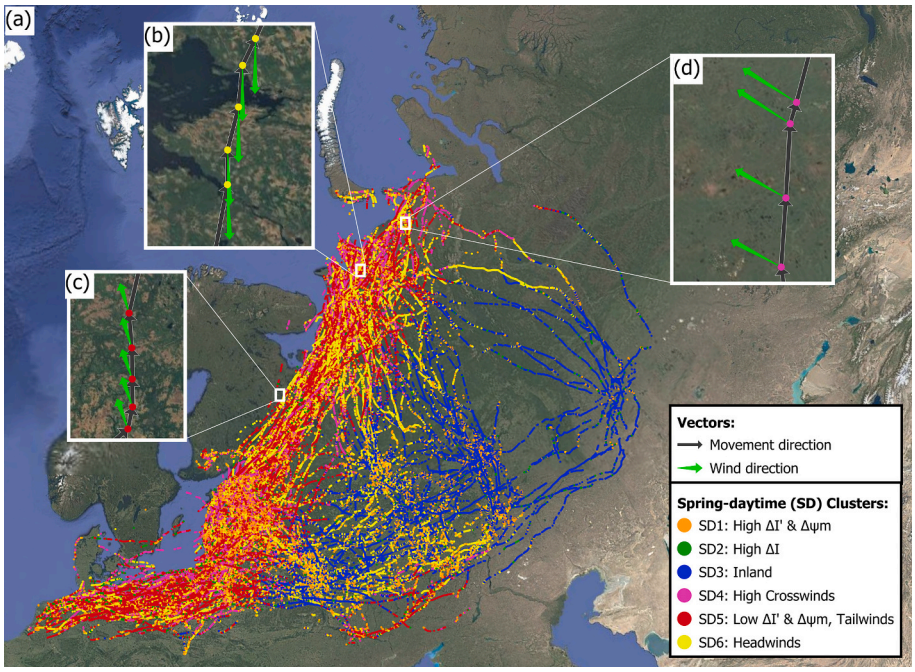


Fig. 10. Distribution of clusters for spring daytime migrations of greater white-fronted geese. The main map (a) shows data points colour-coded by cluster ID. Inset maps (b-d) provide zoomed-in views of a single individual's track during different stages of autumn migration, highlighting the interaction between flight direction (black arrows) and wind direction (green arrows). (b) Cluster SD6, where the bird is flying against the wind. (c) Cluster SD5, illustrating the bird flying with tailwinds. (d) Cluster SD4, showing the bird experiencing strong crosswinds. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and Hedenström, 2007). The magnetoclinic route has been particularly highlighted as a potential mechanism for long-distance avian migration and, under certain conditions, may be indistinguishable from a geomagnetic loxodrome (Åkesson and Bianco, 2016). The coincidence of minimal changes in both $\Delta I'$ and $\Delta\psi_m$ in these clusters implies that geese might employ a flexible orientation strategy that integrates elements of both compass mechanisms, rather than adhering strictly to a single method. Alternatively, the cluster's efficiency might relate to other mechanisms, such as wind-driven odour cues noted in prior studies (Cardé and Willis, 2008; Wallraff, 2004), though we lacked the data to assess olfactory contributions.

The prominence of this cluster's navigational efficiency was most apparent during spring daytime migrations, in which it achieved the lowest deviations from both the final destination and stopover sites. The ability to maintain such precise navigation suggests a strategic use of geomagnetic cues in conjunction with wind assistance to minimise travel time and energy expenditure. The cluster's role in short-term navigation became even more evident when examining deviations from stopovers across different seasons and times of day. During autumn days, for instance, the "Low $\Delta I'$ & $\Delta\psi_m$, Tailwinds" cluster exhibited significantly lower deviations from stopover sites compared to other clusters, even though its advantage in deviations from the final destination was less pronounced. Similarly, during spring nights, this cluster continued to show the strongest alignment with stopover locations, suggesting that geese may strategically prioritise leveraging favourable winds to reach critical stopovers, even if this involves slight deviations from their ultimate destination. This behaviour aligns with the concept of optimal orientation, where birds effectively use wind patterns to minimise travel time and energy expenditure (McLaren et al., 2014; Thorne et al., 2023), yielding direct, goal-oriented flights with minimal deviation from their intended paths. However, other clusters exhibited distinct behaviours, highlighting the need for navigational adjustments in response to different environmental factors.

Clusters exhibiting high changes in I' and ψ_m, often associated with lower speeds and increased turning angles, suggested that the birds were engaging in navigational adjustments. These clusters also showed a wider distribution of headings, suggesting more course corrections compared to the direct flights in other clusters. Importantly, these pronounced turning behaviours were not simply reactions to headwinds, as other clusters with stronger headwinds had lower turning angles. Instead, these adjustments may represent responses to local geomagnetic conditions or other factors prompting directional changes. As an alternative explanation, this behaviour may align with the "trial-and-error" hypothesis (Benhamou, 1997), where birds actively test different headings and assess their progress based on perceived magnetic cues, thereby fine-tuning their navigation.

Clusters characterised by high ΔI may represent another form of recalibration or navigational adjustments. Geese within these clusters experienced significantly elevated ΔI values and notable fluctuations in intensity (F), despite its exclusion from the clustering analysis. These pronounced changes could not be attributed to solar-induced disturbances due to consistently low Kp indices—a measure of geomagnetic activity influenced by solar winds. One possibility is that geese are navigating through areas with strong geomagnetic anomalies. Birds could be using these anomalies as navigational landmarks or "signposts" to confirm their position and adjust their course (Heyers et al., 2017; Wiltschko, 2012; Wiltschko et al., 2006).

A general pattern in the geomagnetic inclination is that its absolute value increases (i.e., the field vector becomes steeper) towards the poles and decreases (i.e., the field gets flatter) towards the equator. Because of this, geese may adjust their flight paths to maintain a consistent or expected rate of change in inclination, thereby tracking their progress along the north-south axis of their migratory route (Kishkinev et al., 2021; Lohmann and Lohmann, 1994). This hypothesis aligns with the greater prevalence of "High ΔI" clusters during spring migration when geese make a pronounced northward movement after an eastward detour. In contrast, fewer latitudinal shifts during autumn could explain the lower prevalence of these clusters, suggesting magnetic anomalies may play a more prominent role in autumn.

As mentioned in the introduction, it has been proposed that birds

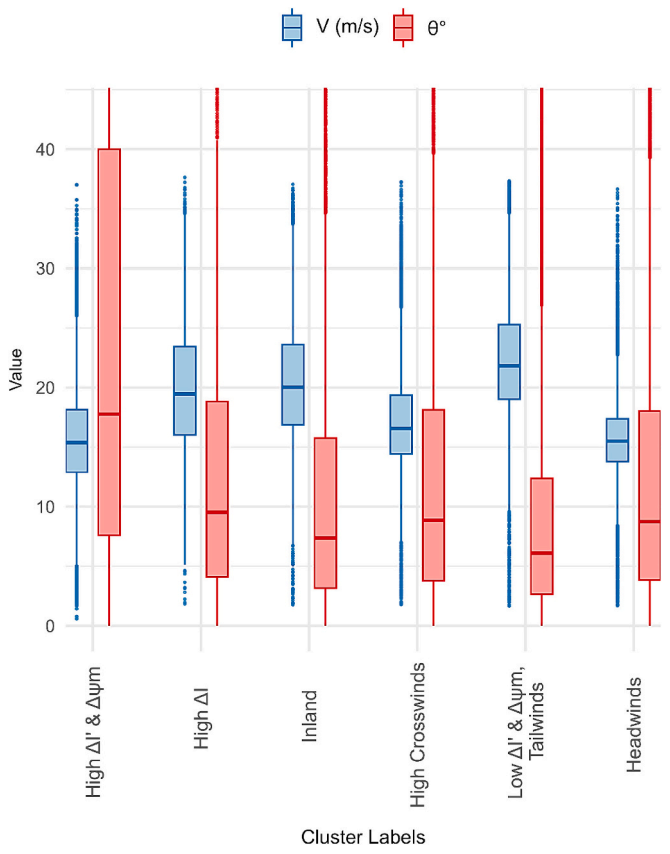


Fig. 11. Boxplots illustrating the distribution of movement speed (V, in blue) and absolute turning angle (θ, in red) for each cluster within spring daytime migration. The y-axis is truncated for visual clarity, and outliers beyond the displayed range are not shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

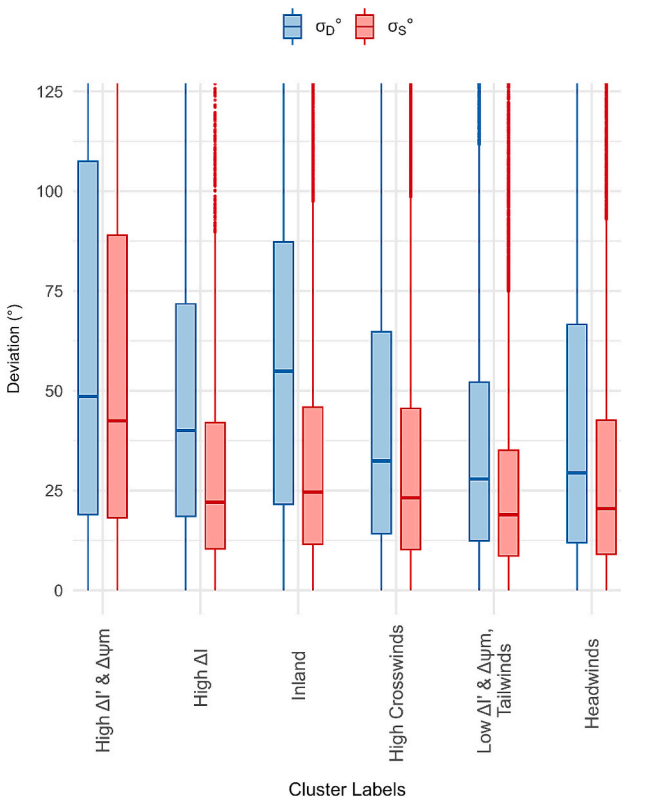


Fig. 12. Boxplots illustrating the distribution of deviations from destinations (σ_D) and stopovers (σ_S), in degrees, for each cluster within the spring daytime migration. The y-axis is truncated for visual clarity, and outliers beyond the displayed range are not shown.

could use these gradients to create a mental map for navigation (Holland, 2014; Kishkinev et al., 2021). The high changes in inclination and intensity observed in our "High ΔI" clusters could be interpreted that the geese were experiencing significant changes in their position on this hypothetical gradient map. However, the high deviations from their destinations observed in these clusters raise questions about whether they truly represent periods of map-based navigation. This discrepancy between the expected behaviour for map-based navigation (low deviations) and the observed behaviour (high deviations) underscores the ongoing debate in the literature regarding the feasibility of birds using geomagnetic maps for long-distance navigation (Boström et al., 2012; Hagstrum, 2023). Further research is needed to resolve these possibilities and clarify the precise role of geomagnetic gradients in the geese's navigational strategies.

Beyond geomagnetic cues, our findings reveal that diurnal and visual landmarks play a complementary role in guiding migratory behaviour. The observed eastward orientation in spring and westward orientation in autumn during nighttime migrations suggests that diurnal cues may help calibrate the geomagnetic compass. This shift in headings and higher deviations from the great circle routes during nighttime indicate that visual cues like the sun's position and landmarks are crucial for maintaining a precise migratory course, supporting the light-dependent nature of the geomagnetic orientation (Able and Able, 1993; Cochran et al., 2004; Muheim et al., 2006). Furthermore, the substantial increase in wind support during spring nighttime migrations, compared to daytime, suggests a reduced reliance on active navigation when visual cues are limited, with birds potentially prioritising energy conservation through greater alignment with the wind (Gulson-Castillo et al., 2023), even if this results in greater deviations from the direct path.

Proximity to coastlines may aid short-term navigation in greater white-fronted geese, particularly during autumn daytime migrations. The "Low ΔI' & $\Delta\psi_m$, Tailwinds" cluster had the shortest distance to the coast and demonstrated significantly lower deviations from stopovers. This indicates that geese might supplement their geomagnetic compass with coastal landmarks when navigating to intermediate targets (Åkesson, 1993; Hedenström and Åkesson, 2016). The greater observed distance from the coast during autumn nighttime compared to daytime further supports that geese rely more on coastlines when landmarks are visible during daylight. It is also possible that geese detect infrasound signals emanating from surf noise in coastal areas (Hagstrum, 2000; Hagstrum, 2023; Patrick et al., 2021); however, testing this possibility lies beyond the scope of our study. The inconsistent link between coastal proximity and navigation precision suggests coastlines might be one of many cues in the geese's multi-modal navigation strategy, rather than the primary one.

While our study provides valuable insights into the navigational strategies of greater white-fronted geese, certain methodological considerations should be acknowledged. The migratory behaviours captured in our dataset reflected a continuum of context-dependent adjustments, often blurring the boundaries between distinct navigational clusters. This interconnectedness likely mirrors the complex, multi-modal approach geese employ, integrating geomagnetic, wind, and visual cues. Such complexity poses challenges for cluster-validation methods like the Silhouette and Calinski-Harabasz indices, which assume clear separations between groups. To mitigate this, we implemented a dendrogram-cutting threshold of 25 %, selected after systematically exploring alternative thresholds. Higher thresholds produced a single dominant cluster, limiting interpretability, whereas lower thresholds increased cluster fragmentation without altering core findings, confirming the robustness of our results. Furthermore, hierarchical

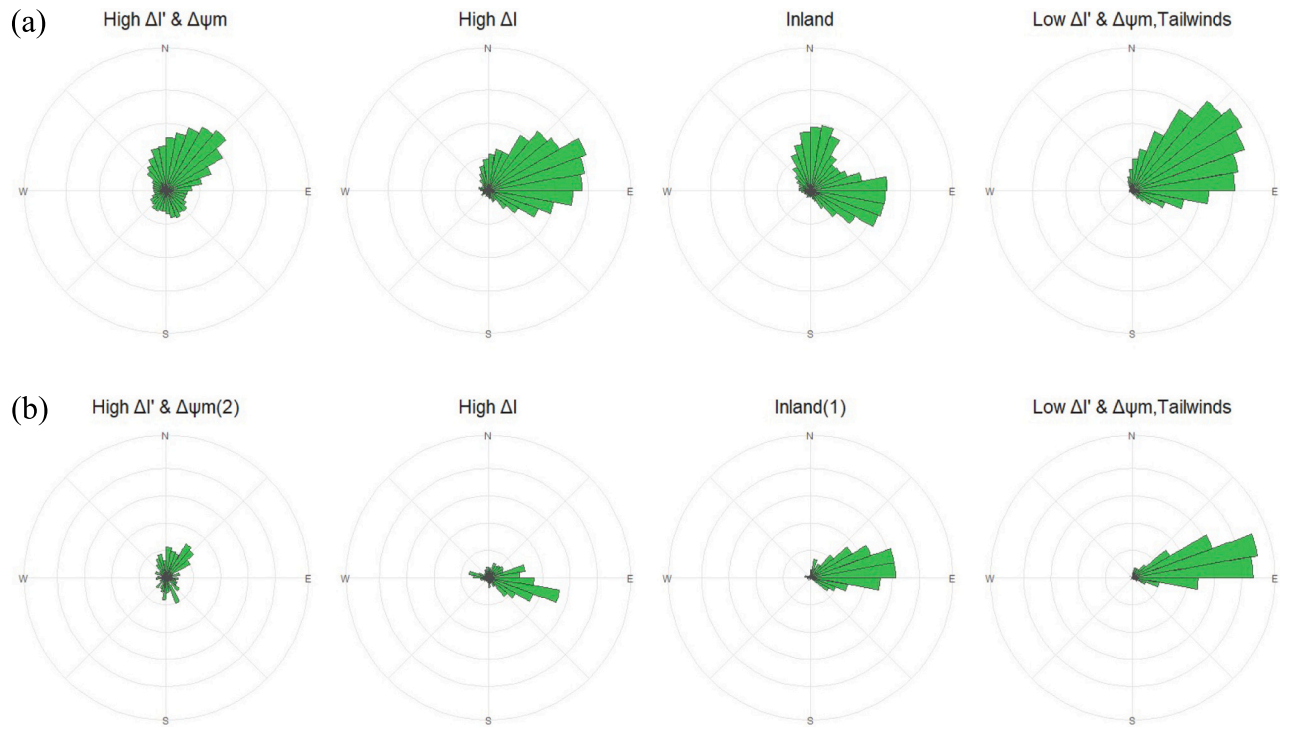


Fig. 13. Circular histograms illustrating the distribution of headings for selected clusters during (a) spring daytime and (b) spring nighttime migrations of greater white-fronted geese.

clustering has particular limitations for hypothesis testing. First, clusters reflect the predefined variables selected (e.g., geomagnetic, wind), which overemphasise measurable cues compared to untracked factors like olfaction. Second, hierarchical clustering assumes equal contribution of all input variables, which might not accurately reflect their true biological importance. Finally, clustering identifies associations but cannot establish causal mechanisms or definitively test competing hypotheses. Therefore, it should be viewed as a hypothesis-generating approach, providing exploratory insights that invite further hypothesis-driven validation.

Furthermore, our study deliberately excluded incomplete migratory tracks to ensure accurate analysis of navigational precision (e.g., deviations from destinations, σ_D), as incomplete journeys lack clear endpoints. Although this may under-represent atypical migrations, such cases were rare and had minimal impact on overall population-level patterns. Additionally, certain individuals were disproportionately represented in specific clusters, likely reflecting individual-specific behaviours, traditional migratory routes, or variations in data coverage. While the clustering process itself remains unaffected because it identifies groupings of data points in attribute space, this underscores that our findings primarily reflect population-level patterns rather than individual-level behaviours. Moreover, the use of global environmental datasets introduces uncertainty; for instance, ERA5 provides interpolated wind estimates that may be less accurate over complex terrain. Nevertheless, such datasets remain the best available for global-scale environmental reconstruction.

For future research, a shift towards trip-level analysis could offer a more comprehensive view of migratory strategies, complementing the point-level analysis used here. Investigating entire migratory trips—from origin to stopovers, between stopovers, and from stopover to destination—may reveal broader patterns and provide insight into how birds adjust their navigation strategies over longer distances and time scales. Future studies could also incorporate additional navigational inputs, such as solar and celestial navigation, olfaction, polarised light, detailed visual landmarks and social cues, to better capture the multisensory integration involved in avian navigation. Lastly, integrating GPS tracking data with physiological or neurobiological measurements would allow for a more nuanced understanding of the internal processes guiding avian migration, revealing how cognitive states influence decision-making during these long journeys.

6. Conclusion

This data mining investigation provides novel insights into the real-time decision-making processes involved in long-distance migration, highlighting the importance of environmental context for migratory strategies. By integrating high-resolution tracking data with diverse environmental cues, our study revealed a dynamic interplay between geomagnetic, wind, and visual information in shaping the migratory strategies of greater white-fronted geese. Our study provides empirical support for the use of geomagnetic loxodrome and magnetoclinic routes in free-flying birds within their natural environment. Under favourable tailwinds, geese employ flexible strategies that adapt to environmental conditions, diurnal cycles, and likely flock-specific characteristics, facilitating efficient flights either towards stopovers or simultaneously aligning with stopovers and final destinations. Wind emerged as a critical factor in geese's navigation, with tailwinds enabling efficient, energy-saving travel and crosswinds necessitating compass recalibration to minimise drift. These findings, along with the observed integration of visual cues with geomagnetic information, underscore the complexity and adaptability of avian navigation, extending beyond the traditional singular map-and-compass model.

CRediT authorship contribution statement

Ali Moayedi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jed A. Long:** Writing – review & editing, Validation, Supervision, Methodology. **Andrea Kölzsch:** Writing – review & editing, Resources, Data curation. **Helmut Kruckenberg:** Writing – review & editing, Resources. **Fernando Benitez-Paez:** Writing – review & editing, Supervision, Software. **Urška**

Demšar: Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

Ethical considerations

All data collection procedures were carried out in accordance with ethical guidelines of the respective countries. Catching and equipment of birds with transmitters were carried out in Russia under the umbrella of a personal permit to Petr Glazov at the Institute of Geography of the Russian Academy of Sciences, in the Netherlands with approval by the Animal Welfare Committee of the Royal Netherlands Academy of Arts and Sciences (DEC NIOO13.14) and in Germany with the permission of the Lower Saxony State Office for Consumer Protection and Food Safety (LAVES; Akt.Z. 33.19–4202-04-15/1956).

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests (Andrea Kolzsch reports financial support was provided by the Dutch Society of Goose catchers. Andrea Kolzsch reports financial support was provided by Lower Saxony Ministry of Food, Agriculture and Consumer Protection. Andrea Kolzsch reports financial support was provided by German Aerospace Center. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.)

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecoinf.2025.103218.

Data availability

The processed GPS-tracking dataset is available on Zenodo (DOI: https://doi.org/10.5281/zenodo.15458188; CC-BY 4.0). The complete analysis workflow (R and Python scripts) is archived on Zenodo (DOI: https://doi.org/10.5281/zenodo.15358009; MIT licence) and mirrored on GitHub (github.com/BEGIN-StAndrews/geese_migration_analysis). Raw GPS data originate from five Movebank studies (IDs 127892189, 408961322, 13183695, 180156318, 44083081) and are available under their original terms.

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